

Pautlen

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Assignment 5 - Compiler

This project is about a simple compiler for 'alfa' language. 'Alfa' was thought for educational purposes, so this will not be such interesting to consider as a real option language - don't waste time here!

Despite previous line, it's been interesting to have a general idea about compilers stuff.

Makefile

For help, run:

make help

```
> make help
all                Produces compiler
clean              Cleans output files
default            Equivalent to 'make all'
help               This guide
```

Important: **y.output** is stored at **root/info/y.output**.

Compiler Info

For running compiler, before you must have it:

```
> make
```

Now, you can use it as follows:

```
> ./alfa <source.alf> <dest.nasm>
```

For compiling resulting *nasm* file:

```
> ./produce.sh <nasm_file> <program_name>
```

Finally, for executing your alfa program:

```
> ./<program_name>
```

If you want to launch compiler with **DEBUG** mode - it'll print traces, rules applied, etc - you must uncomment this at **alfa.h**:

```
97 /**
98  * Prints failure trace
99  */
100 // #define _DEBUG_ <----- HERE!
101 #ifdef _DEBUG_
102 #define TRACE eprintf("Trace coming from:\t @ file: %s @ funct: %s @
line: >
103 #else
104 #define TRACE
105 #endif
```

Comments about design

A common header - *alfa.h* - contains all common stuff (buffers & related) used by *alfa.c* - launcher -, *alfa.y* - bison - and *alfa.l* - flex.

The project uses the next data structures:

- Hash, with these features:
 - Auto incremental size.
 - Does **not** support deletions.
 - Linea probing.
 - Keys are whatever is desired. *Hashcode* prototype is at **root/inc/prototypes.h**.
- Symbols:
 - The different options alfa language contain. Actually, as we've designed them is not as we though at the first the moment, but deadlines and stuff overload forced us to end them as quickly as possible.
- Symbols Table:
 - Uses two Hash tables to store symbols, depending on the scope - context.
- Generator:
 - A *Netwide Assembler* generator library, used by *alfa.y* - bison - to produce final code which'll be compiled as assembler to get final desired program.